

# **316-211 Ceramic materials**

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# ***Objectives***

1. To explain basic principles in terms of firing and kilns
2. To classify ceramic bodies
3. To explain basic principles of structural ceramics

**1. เตาเผาและอิทธิพลของความร้อนที่มีต่อวัตถุดิบ**  
(Kiln and influence of heat on raw materials)

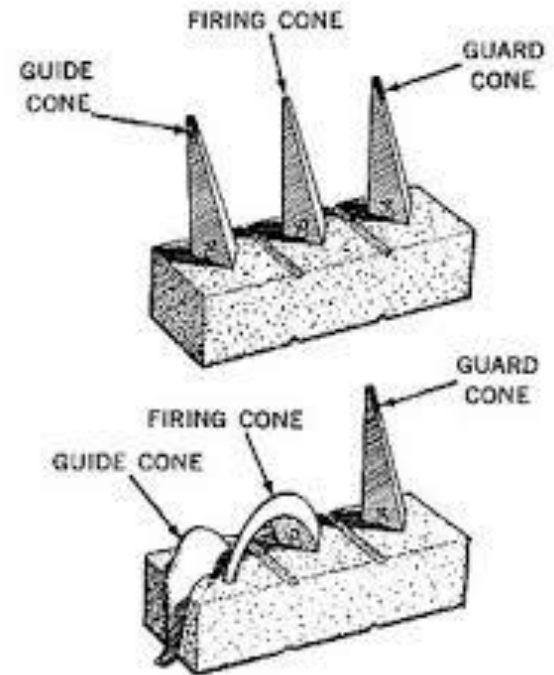
## **How Heat is Transferred**

Heat moves through the kiln from hotter to cooler zones by:

1. convection
2. conduction
3. radiation

# Convection

Convection is the first step in the heating process in the kiln. Air is heated as it passes across the warming kiln elements. As the hot air rises and cool air falls, air currents are created which circulate hot air to cooler places in kiln. This heat is transferred to the ware, shelves, etc.



What is seger cone temp chart?

The kiln will not be uniform in temperature at this early stage of firing unless the hot air is pushed through the kiln by mechanical means. Low cone firings such as 022 and 021 depend more heavily on convection for heat transfer.

The most common type of convection we are familiar with is wind chill. The cool air passes across the face and pulls heat from our warmer body, which lowers our skin temperature.

## **Conduction**

When heat moves through a solid, it is conducted. An example would be heat moving through the handle of a saucepan. This is a slow way to heat, but the handle will eventually get hot.

In a kiln, conduction moves heat from the inside to the outside of the kiln and from the outside to the inside of the ware. Conduction is the main way we get uniform heating in the kiln. This is a slow process and if we fire too fast, the inside of our ware will receive too little heat and not fire properly.

## **Radiation**

At the beginning of the firing, the elements are the hottest part of the kiln. The heat from the elements radiates out - like the sun warming us on a cool day. Eventually the firebrick and the ware will also get hot and will radiate heat as well.

As the temperature increases, more and more of the heat is transferred by radiation from the heating elements. For uniform heating, is important that all surfaces of the ware be exposed to heating elements, even partially.

# Firing atmosphere

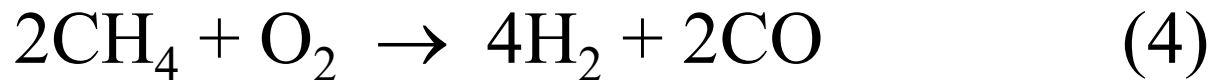
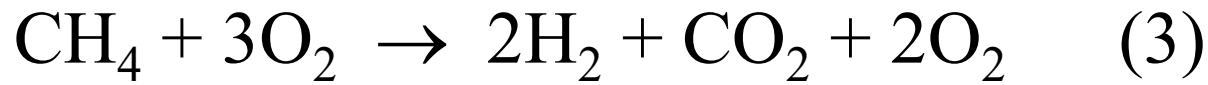
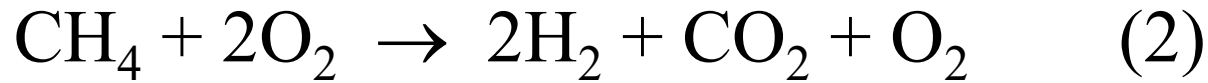
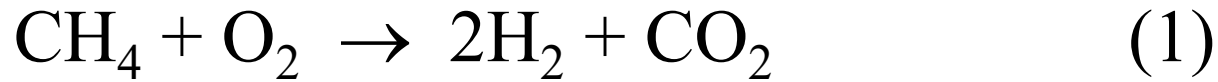
## Reduction Firing

In the reduction process the ceramist reduces the amount of oxygen allowed to enter the firing chamber. A fuel-burning kiln demands a great deal of oxygen: it is very natural for the atmosphere inside a fuel-burning kiln to become depleted of oxygen during the firing.

Reduction leaves its mark on both clay bodies and glazes. It modifies color and visual texture. Clay body color is deepened, sometimes moving to rich oranges and reds and sometimes to gray colors.

Glaze color can be strongly marked by reduction: for example, copper will turn a blood red, white glazes take on a cream color with a broken texture of dark spots, iron greens and ocher colors become burnt oranges and brick reds; sky blues become slate blues.

## Possible reactions during combustion





Oxidation  $2^+ \rightarrow$  Oxidation 0

Melting point of PbO =  $888^\circ\text{C}$

Melting point of Pb metal =  $327^\circ\text{C}$

Oxidation number

Fe = 0

FeO =  $2^+$

Fe<sub>2</sub>O<sub>3</sub> =  $3^+$

Fe<sub>3</sub>O<sub>4</sub> =  $2^+$  and  $3^+$

# Low-Fire Reduction

## A Sawdust Firing



Applying a stain to the surface of the piece.



Placing the piece on a bed of sawdust. More sawdust will be added to cover the work.



Placing wadded up newspapers on the sawdust to start the fire.



Setting the newspaper on fire.



Covering the fire with sheets of metal after the sawdust has been ignited.



Removing the cooled work a few hours after the four to six hour firing is finished. The darkened work can now be cleaned and waxed.

# Oxidation Firing

In this kind of firing oxygen is allowed free access to the kiln chamber. In the past this was not so easy, wood-fired kilns naturally went into reduction during the firing.

Even then not all ceramists fired in reduction. For example, lead glazes boil and bubble in reduction. Ceramists who finished their pieces with lead glazes took the trouble to control their kiln firings and avoid reduction. With the advent of modern kilns it became very easy to fire in oxidation. Fuel-burning kilns, whose burners are fan driven, lend themselves to oxidation firing.

The contemporary ceramist must decide whether to use an oxidation or reduction firing atmosphere. This will dictate the choice of a kiln. Neither oxidation nor reduction is superior; both are tools to be used by the ceramist when appropriate.

# Fuel-Burning Kilns

Fuels can be divided into solid, liquid, or gaseous. Until the late 19th century only solid fuels were available. Animal dung, wood, and coal are all solid fuels. In kilns fired with solid fuels the unburned ash residue must constantly be removed. Only one solid fuel - wood - finds a great deal of use among contemporary ceramists in the developed countries. Now most fuel-burning kilns are fueled with a liquid such as oil, or a gaseous fuel such as natural gas or propane.

# Kilns

Kilns are basically containers for heat. In the most primitive forms, this could be nothing more than a trench or pit dug into the earth. Today, there are many different types of kilns. Here is an introduction to some of the most common types used for pottery and the ceramic arts.

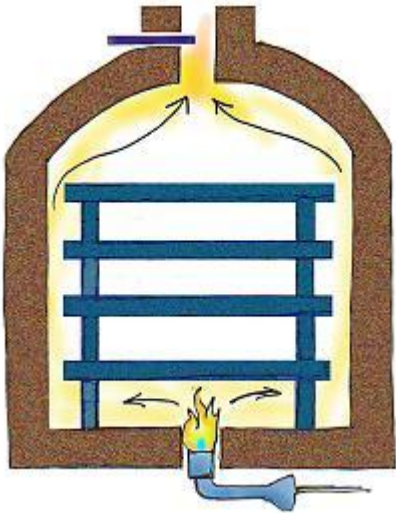
There are two basic types of kilns that all other types fall into. These two primary types are continuous (tunnel) kilns and periodic (intermittent) kilns.

**Continuous kilns** are always firing; they never cool. The ware to be fired is loaded into cars and slowly goes through the kiln on a track or rail. After cooling, the still-moving cars are unloaded and reloaded to begin the circuit again. These kilns are used in industry.

**Periodic kilns** are the ones most familiar to potters. They are fired on an intermittent schedule; the kilns are loaded, brought to temperature, cooled, then unloaded. The kiln is not firing all the time.

## Updraft Kilns

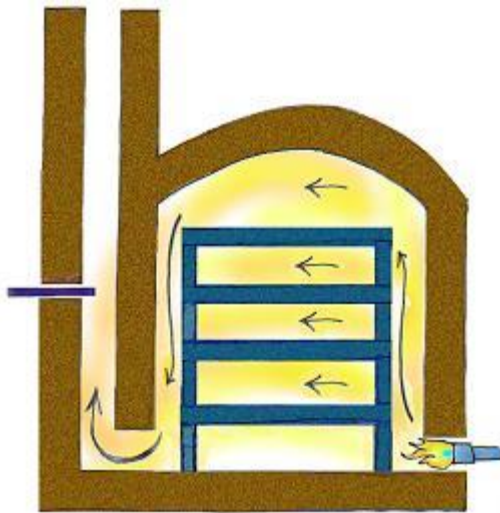
Updraft kilns are those in which the flame is introduced into the bottom of the kiln, at or below floor level, and exhausted out the top. Updraft kilns consist of three basic components: the firebox, the damper, and the stack area.



The firebox is where the flame enters. The damper is at the top of the kiln and controls the exhaust (and by association, the kiln's atmosphere). The stack area is where the pots are set and is between the firebox and damper.

Although an updraft kiln tends to be less fuel efficient than a downdraft kiln, most commercially built fuel-burning kilns are updrafts. This is mainly due to their simplicity to build, pack, and ship.

# Downdraft Kilns



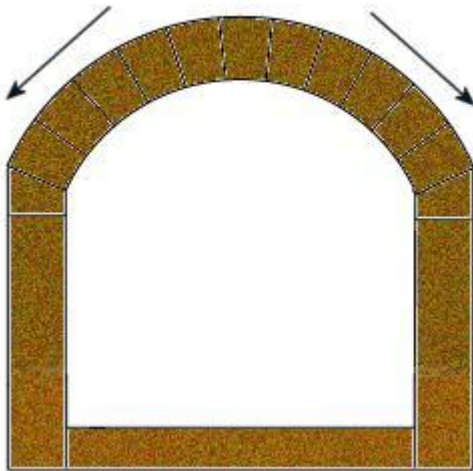
Downdraft kilns are designed to force the flame and heated air to circulate through the kiln. Flame is introduced at the bottom of the kiln and naturally flows upward. The construction forces the flame back downward, to exhaust at the bottom of the kiln.

Downdraft kilns consist of four main components: the firebox, the stack area, the damper, and the chimney. The addition of the chimney helps create draw, or air flow.

Sprung arch downdraft kilns are probably the most commonly potter-built kiln type in the United States

## Sprung Arch Kilns

The term "sprung arch" refers to the roof structure of the kiln. The arch is built using a form; once completed, the form is removed and the arch will stay in place without further interior support.

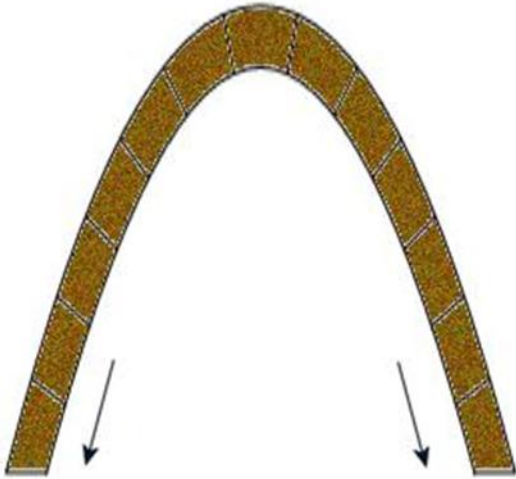


Sprung arches do, however, displace weight outward. Due to this outward tension, sprung arch kilns are framed with angle iron to support the arch and reduce any movement of the firebricks used to build the kiln.

Sprung arch kilns may be designed as either updraft or downdraft kilns.

## Catenary Arch Kilns

Catenary arches, once built, are self-supporting. The arch itself is an upside-down replica of the curve produced when a heavy rope or chain is hung from two points on the same horizontal plane. Like sprung arches, catenary arches must be supported by a form while they are being constructed.



Catenary arch kilns are often quite beautiful; it is a graceful form. However, the interior does not lend itself to an efficient use of space in the stack area. Like sprung arch kilns, a catenary arch kiln may be either an updraft or downdraft kiln, depending on how the air and flame flows through the kiln.

## **Types of Kiln Shelves**

Kiln shelves are made from highly refractory materials.

High-alumina shelves: Dry-pressed high-alumina shelves are tougher and last longer than slip-cast high-alumina shelves. Generally speaking, high-alumina shelves are best used in electric kilns. They especially should not be used for raku, salt, or wood firings.

Silicon carbide shelves: Silicon carbide is a conductor; I strongly recommend they never be used in electric kilns. If silicon carbide shelves are used in an electric kiln, the kiln should be disconnected to all power before the shelves are placed or removed.

# Refractory

A refractory material is one that retains its strength at high temperatures. ASTM C71 defines refractories as "non-metallic materials having those chemical and physical properties that make them applicable for structures, or as components of systems, that are exposed to environments above 1,000 °F (811 K; 538 °C)".

# Refractory materials

Refractory materials must be chemically and physically stable at high temperatures. Depending on the operating environment, they need to be resistant to thermal shock, be chemically inert, and/or have specific ranges of thermal conductivity and of the coefficient of thermal expansion.

The oxides of aluminium (alumina), silicon (silica) and magnesium (magnesia) are the most important materials used in the manufacturing of refractories. Another oxide usually found in refractories is the oxide of calcium (lime). Fire clays are also widely used in the manufacture of refractories.

Refractories must be chosen according to the conditions they will face. Some applications require special refractory materials. Zirconia is used when the material must withstand extremely high temperatures. Silicon carbide and carbon (graphite) are two other refractory materials used in some very severe temperature conditions, but they cannot be used in contact with oxygen, as they will oxidize and burn.

# Classification of refractory materials

Refractories can be classified on the basis of chemical composition, method of manufacture, physical form or according to their applications, fusion temperature.

## Based on chemical composition

### Acidic refractories

Acidic refractories consist of mostly acidic materials like alumina ( $\text{Al}_2\text{O}_3$ ) and silica ( $\text{SiO}_2$ ). They are generally not attacked or affected by acidic materials, but easily affected by basic materials. They include substances such as silica, alumina, and fire clay brick refractories. Notable reagents that can attack both alumina and silica are hydrofluoric acid, phosphoric acid, and fluorinated gases (e.g.  $\text{HF}$ ,  $\text{F}_2$ ). At high temperatures, acidic refractories may also react with limes and basic oxides.

## Neutral refractories

These are used in areas where slags and atmosphere are either acidic or basic and are chemically stable to both acids and bases. The main raw materials belong to, but are not confined to, the  $R_2O_3$  group. Common examples of these materials are alumina ( $Al_2O_3$ ), chromia ( $Cr_2O_3$ ) and carbon.

## Basic refractories

These are used in areas where slags and atmosphere are basic; they are stable to alkaline materials but could react with acids. The main raw materials belong to the  $RO$  group to which magnesia ( $MgO$ ) is a very common example. Other examples include dolomite and chrome-magnesia. For the first half of the twentieth century, the steel making process used artificial periclase (roasted magnesite) as a lining material for the furnace.



# Ceramic fiber

High-temperature insulation wool (HTIW), known as ceramic fiber wool until the 1990s, is one of several types of synthetic mineral wool, generally defined as those resistant to temperatures above 1000°C. The first variety, aluminium silicate fibre, developed in the 1950s, was referred to as refractory ceramic fibre.

Due to the costly production, and limited availability compared to mineral wool, HTIW products are almost only used in high temperature industrial applications and processes.

Comparison of the mass for the different wall linings

Heavyweight lining: 1500–3500 kg/m<sup>3</sup>,

Lightweight lining: 500–1000 kg/m<sup>3</sup>,

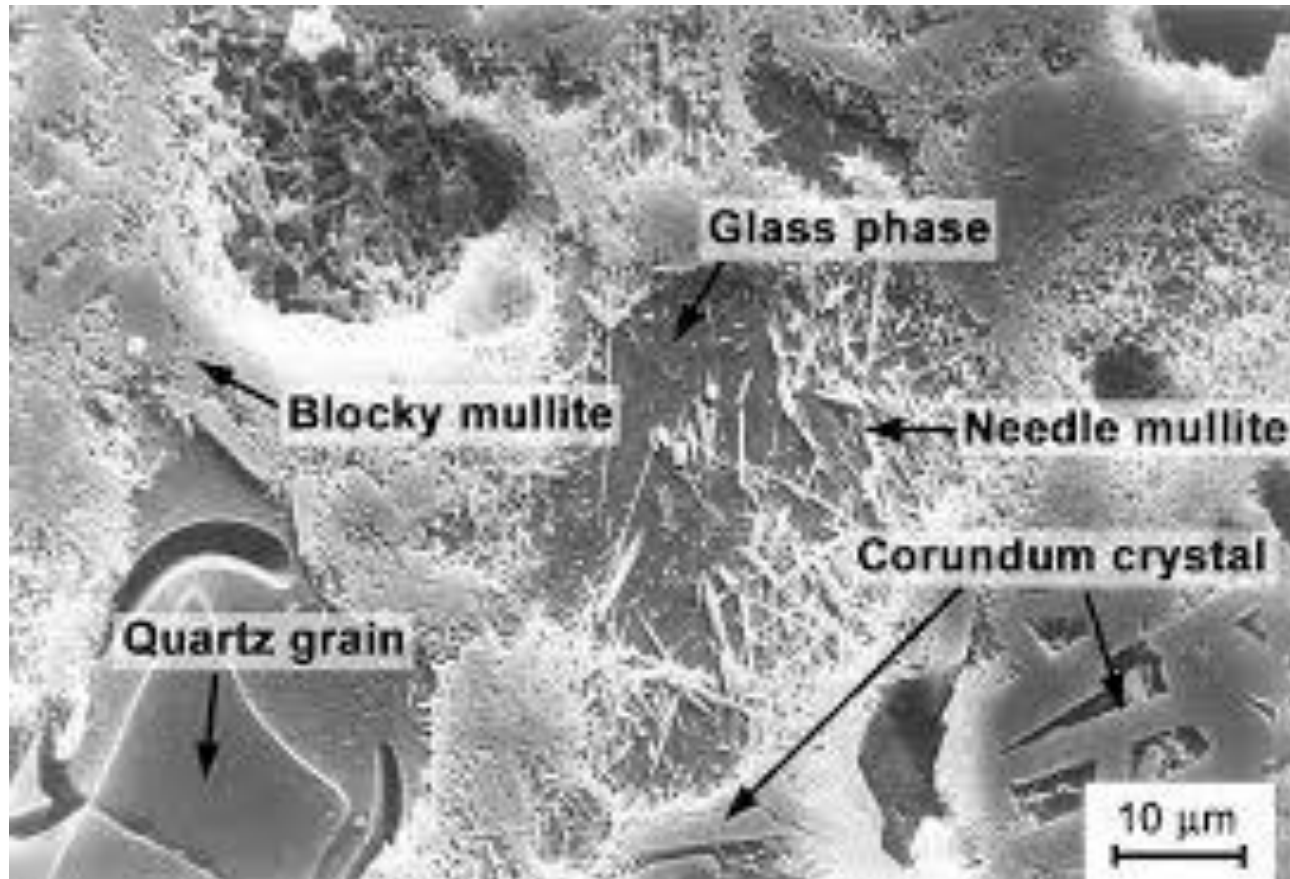
Lining with HTIW: 160–300 kg/m<sup>3</sup>.



# How Temperature Changes Ceramic bodies

| Temperature               | Stage of Firing                                                                                                |
|---------------------------|----------------------------------------------------------------------------------------------------------------|
| 212°F/100°C               | Water boils causing wet greenware to explode.                                                                  |
| 428°F/220°C               | During cooling if cristobalite crystals are present in clay they rapidly shrink 3% potentially cracking ware.  |
| 662°F–932°F/350°C–500°C   | Permanent dehydration of clay occurs. Clay is chemically changed.                                              |
| 1063°F/573°C              | Quartz inversion happens causing quartz in clay and glazes to expand when heating and contract during cooling. |
| 1112°F–1652°F/600°C–900°C | Organic and inorganic matter is being burned off out of the clay.                                              |
| 2012°F/1100°C             | Mullite crystals begin forming in porcelain clays.                                                             |

# Microstructure of porcelain body



# Types of Ceramic Wares

- Object is air dried — **green ware**
- First firing — **biscuit firing (bisque)**
- Subsequent firings — **glazed ware**

## Stages in Firing of Clay Bodies

### Stage 1 — **Drying**

- **rt** —————→ **250°C**
- **residual water** is removed
- Done at slow rate to prevent explosive vaporization

## Stage 2 — Dehydroxylation and Oxidation

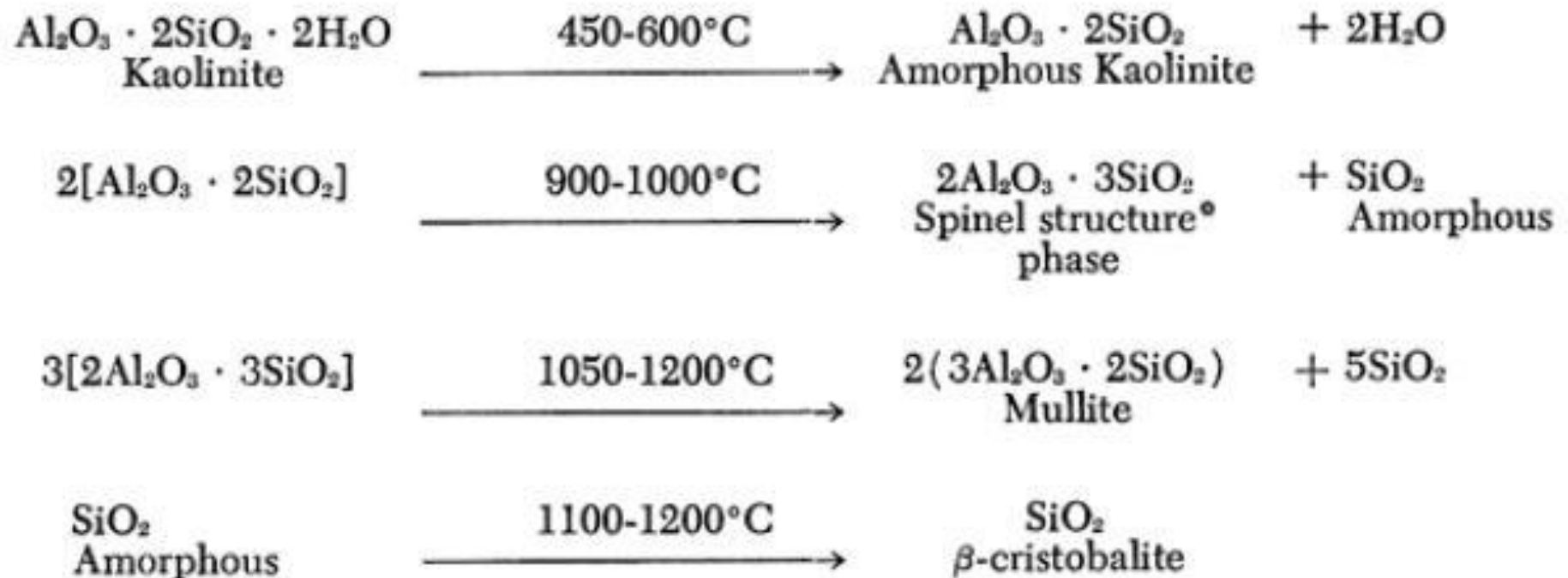
- **250°C**  $\longrightarrow$  **500°C**
- **lattice** or chemically bound water is removed
- organic matter burns off

## Stage 3 — Vitrification

- **850°C**  $\longrightarrow$  **1450°C**
- fluxes melt
- glassy phase develops
- clay undergoes various crystalline phase changes

### **Stage 4 — Cooling**

- back to room temperature
- crystalline inversions can occur and ceramic can crack



earthenware — 1150°C

bone china — 1250°C

hard porcelain — 1450°C

## 2. ชนิดของเนื้อดินปั้นเซรามิก (Types of ceramic bodies)

### Types of Ceramic Pottery

- Terra-cotta — flower pots
- Earthenware — highly glazed, red body
- Stoneware — dinner dishes
- Bone china — fine china
- Porcelain — toilets and sinks
- Vitrified Pottery

# Earthenware

Earthenware is glazed or unglazed nonvitreous pottery which has normally been fired below 1200°C. Porcelain, bone china and stoneware, all fired at high enough temperatures to vitrify, are the main other important types of pottery.

Earthenware comprises all primitive pottery whatever the color, all terra-cottas, most building bricks.



Earthenware, when fired, is opaque and non-vitreous, soft and capable of being scratched with a knife. The Combined Nomenclature of the European Communities describes it as being made of selected clays sometimes mixed with feldspars and varying amounts of other minerals and white or light-colored (i.e., slightly greyish, cream or ivory).



## Characteristics

Generally, earthenware bodies exhibit higher plasticity than most whiteware bodies and hence are easier to shape by RAM press, roller-head or potter's wheel than bone china or porcelain.

Due to its porosity, earthenware, with a water absorption of 5-8%, must be glazed to be watertight. Earthenware has lower mechanical strength than bone china, porcelain or stoneware, and consequently articles are commonly made in thicker cross-section, although they are still more easily chipped.

Darker-colored "terracotta" earthenwares, typically orange or red, due to a comparatively high content of iron oxide are widely used for flower pots, tiles and some decorative and oven wares.

## Production

A general body formulation for contemporary earthenware is 25% kaolin, 25% ball clay, 35% quartz and 15% feldspar.

Earthenware may be biscuit (or "bisque") fired to temperatures between 1,000 and 1,150 °C (1,800 and 2,100 °F) and glost-fired (or "glaze-fired") to between 950 to 1,050 °C (1,740 to 1,920 °F), the usual practice in factories and some studio potteries. Some studio potters follow the reverse practice, with a low-temperature bisque firing and a high-temperature glost firing. The firing schedule will be determined by the raw materials used and the desired characteristics of the finished ware.

After firing most earthenware bodies will be colored white, buff or red. For red earthenware, the firing temperature affects the color of the clay body. Lower temperatures produce a typical red terracotta color; higher temperatures will make the clay brown or even black. Higher firing temperatures may cause earthenware to bloat.

# Stoneware

Stoneware is a rather broad term for pottery or other ceramics fired at a relatively high temperature. A modern technical definition is a vitreous or semi-vitreous ceramic made primarily from stoneware clay or non-refractory fire clay. Whether vitrified or not, it is nonporous (does not soak up liquids); it may or may not be glazed. Historically, across the world, it has usually been developed after earthenware and before porcelain, and has often been used for high-quality as well as utilitarian wares.



"Stoneware, which, though dense, impermeable and hard enough to resist scratching by a steel point, differs from porcelain because it is more opaque, and normally only partially vitrified. It may be vitreous or semi-vitreous. It is usually coloured grey or brownish because of impurities in the clay used for its manufacture, and is normally glazed."



Five basic categories of stoneware have been suggested:

***Traditional stoneware*** - a dense and inexpensive body. It is opaque, can be of any colour and breaks with a concoidal or stony fracture.

Traditionally made of fine-grained secondary, plastic clays which can be used to shape very large pieces.

***Fine stoneware*** - made from more carefully selected, prepared, and blended raw materials. It is used to produce tableware and art ware.

***Chemical stoneware*** - used in the chemical industry, and when resistance to chemical attack is needed. Purer raw materials are used than for other stoneware bodies. Ali Baba is a popular name for a large chemical stoneware jars of up to 5,000 litres capacity used to store acids.

***Thermal shock resistant stoneware*** – has additions of certain materials to enhance the thermal shock resistance of the fired body.

***Electrical stoneware*** - historically used for electrical insulators, although it has been replaced by electrical porcelain.

## Materials and firing

The key raw material in stoneware is either naturally occurring stoneware clay or non-refractory fire clay. The mineral kaolinite is present but disordered, and although mica and quartz are present their particle size is very small. Stoneware clay is often accompanied by impurities such as iron or carbon, giving it a "dirty" look, and its plasticity can vary widely. Non-refractory fire clay may be another key raw material. Fire clays are generally considered refractory, because they withstand very high temperatures before melting or crumbling. Refractory fire clays have a high concentration of kaolinite, with lesser amounts of mica and quartz. Non-refractory fire clays, however, have larger amounts of mica and feldspar.

Formulations for stoneware vary considerably, although the vast majority will conform to: plastic fire clays, 0 to 100 percent; ball clays, 0 to 15 percent; quartz, 0 to 30 percent; feldspar and chamotte, 0 to 15 percent.

Stoneware can be once-fired or twice-fired. Maximum firing temperatures can vary significantly, from 1100 °C to 1300 °C depending on the flux content. Typically, temperatures will be between 1180 °C to 1280 °C, the higher end of which equate to Bullers Rings 38 to 40 or Seger cones 4 to 8. To produce a better quality fired glaze finish, twice-firing can be used. This can be especially important for formulations composed of highly carbonaceous clays. For these, biscuit firing is around 900 °C, and glost firing (the firing used to form the glaze over the ware) 1180–1280 °C. Water absorption of stoneware products is less than 1 percent.

Another type, Flintless Stoneware, has also been identified. It is defined in the UK Pottery (Health and Welfare) Special Regulations of 1950 as: "Stoneware, the body of which consists of natural clay to which no flint or quartz or other form of free silica has been added."

Traditional East Asian thinking only classifies pottery into "low-fired" and "high-fired" wares, equating to earthenware and porcelain, without the intermediate European class of stoneware, and the many local types of stoneware were mostly classed as porcelain, though often not white and translucent.

# Porcelain

Porcelain /'pɔːrsəlɪn, 'pɔːrslɪn/ is a ceramic material made by heating materials, generally including kaolin, in a kiln to temperatures between 1,200 and 1,400 °C (2,200 and 2,600 °F). The toughness, strength and translucence of porcelain, relative to other types of pottery, arises mainly from vitrification and the formation of the mineral mullite within the body at these high temperatures.



Porcelain was first developed in China around 2,000 years ago, then slowly spread to other East Asian countries, and finally Europe and the rest of the world. Though its manufacturing process is more demanding than that for earthenware and stoneware, the two other main types of pottery, it has usually been regarded as the most prestigious type of pottery for its delicacy, strength and beauty, and its white colour. It combines well with both glazes and paint, and can be modelled very well, allowing a huge range of decorative treatments in tablewares, vessels and figurines. It also has many uses in technology and industry.

The European name, porcelain in English, come from the old Italian porcellana (cowrie shell) because of its resemblance to the translucent surface of the shell. Porcelain is also referred to as china or fine china in some English-speaking countries, as it was first seen in imports from China. Properties associated with porcelain include low permeability and elasticity; considerable strength, hardness, toughness, whiteness, translucency and resonance; and a high resistance to chemical attack and thermal shock.

Porcelain has been described as being "completely vitrified, hard, impermeable (even before glazing), white or artificially coloured, translucent (except when of considerable thickness), and resonant." However, the term porcelain lacks a universal definition and has "been applied in a very unsystematic fashion to substances of diverse kinds which have only certain surface-qualities in common".

Traditional East Asian thinking only classifies pottery into low-fired wares (earthenware) and high-fired wares (porcelain), without the intermediate European class of stoneware, and the many local types of stoneware were mostly classed as porcelain, though often not white and translucent. Terms such as "porcellaneous" or "near-porcelain" may be used in such cases. A high proportion of modern porcelain is made of the variant bone china.

# Bone china

Bone china is a type of soft-paste porcelain that is composed of bone ash, feldspathic material, and kaolin. It has been defined as ware with a translucent body containing a minimum of 30% of phosphate derived from animal bone and calculated calcium phosphate. Bone china is the strongest of the porcelain or china ceramics, having very high mechanical strength and chip resistance, and is known for its high levels of whiteness and translucency. Its high strength allows it to be produced in thinner cross-sections than other types of porcelain. Like stoneware it is vitrified, but is translucent due to differing mineral properties



## Production

The production of bone china is similar to porcelain, except **more care is needed because of its lower plasticity and a narrower vitrification range**. The traditional formulation for bone china is about 25% kaolin, 25% Cornish stone and 50% bone ash. **The bone ash that is used in bone china is made from cattle bones** that have a lower iron content. These bones are crushed before being degelatinised and then calcined at up to 1250 °C to produce bone ash. The ash is milled to a fine particle size. The kaolin component of the body is needed to give the unfired body plasticity which allows articles to be shaped. This mixture is then fired at around 1200 °C. The raw materials for bone china are comparatively expensive, and the production is labour-intensive, which is why bone china maintains a luxury status and high pricing.

**Bone china consists of two crystalline phases, anorthite ( $\text{CaAl}_2\text{Si}_2\text{O}_8$ ) and  $\beta$ -tricalcium phosphate/whitlockite ( $\text{Ca}_3(\text{PO}_4)_2$ ) embedded in a substantial amount of glass.**

# Ceramic Glaze

Glaze is a special type of glass, made for coating ceramic products. Whereas glass is suitable for forming into bottles or windows, glaze is different because it is applied on a ceramic surface and must form a hard, durable coating after being melted in the kiln. It must not run off the product and must stay on the product after firing without cracking.

## Classification (earthenware, stoneware etc.)

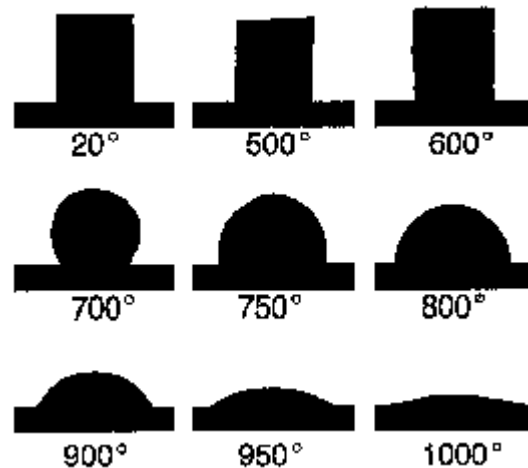
Although there are many different ways to classify glazes, the simplest way to understand them is according to the firing temperature. The useful range of temperatures for glaze melting is from 900°-1300°C. In this book, we talk about two different categories of glaze:

- low temperature from 900-1100°C, called earthenware;
- high temperature from 1100-1300°C, called stoneware.

These two categories are used because they require different raw materials as the main ingredients of the glaze.

The temperature at which melting starts depends on the materials in the glaze. Silica alone melts at 1715°C, but with additions of other materials the melting point will go down. Aluminum oxide ( $\text{Al}_2\text{O}_3$ ) melts at 2050°C and calcium oxide ( $\text{CaO}$ ) at 2570°C, but a mixture of 62% silica, 14.75% aluminum oxide and 23.25% lime melts at only 1170°C. A mixture which has a lower melting point than any of the single materials in the mixture is called an eutectic.

A mixture with many different materials will form eutectics (and will melt) at a lower temperature. Fine grinding of the glaze materials and prolonged firing time above the sintering temperature will also lower the melting point.



**Celadon glaze** refers to a family of usually transparent glazes, many with pronounced "crackle", or tiny cracks in the glaze produced in a wide variety of colors, generally used on porcelain or stoneware clay bodies. Celadon glazes have such popularity and impact that pieces made with it are often referred to as "celadons."

Celadon glazes can be produced in a variety of colors, including white, grey, blue and yellow, depending on several factors: 1) the thickness of the applied glaze, 2) the type of clay to which it is applied, 3) the exact chemical makeup of the glaze, 4) the firing temperature 5) the degree of reduction in the kiln atmosphere and 6) the degree of opacity in the glaze. The exact shade produced often varied between different parts of a single firing in a kiln. The most famous and desired shades range from a very pale green to deep intense green, often meaning to mimic the green shades of jade. The main color effect is produced by iron oxide in the glaze recipe or clay body. Celadons are almost exclusively fired in a reducing atmosphere kiln as the chemical changes in the iron oxide which accompany depriving it of free oxygen are what produce the desired colors. As with most glazes, crazing (a glaze defect) can occur in the glaze and, if the characteristic is desirable, is referred to as "crackle" glaze.



## Elaine's Celadon Base Glaze

(Cone 8–11, reduction)

|                                  |         |
|----------------------------------|---------|
| Whiting .....                    | 21.2 %  |
| Zinc Oxide .....                 | 2.7     |
| Custer Feldspar .....            | 24.9    |
| Frit 3110 (Ferro) .....          | 8.8     |
| EPK (Edgar Plastic Kaolin) ..... | 17.5    |
| Silica (200-mesh Flint) .....    | 24.9    |
|                                  | 100.0 % |

|                 |       |
|-----------------|-------|
| White:          |       |
| Tin Oxide ..... | 0.7 % |

|                        |       |
|------------------------|-------|
| Green:                 |       |
| Mason Stain 6201 ..... | 0.9 % |

|                        |       |
|------------------------|-------|
| Iron Blue:             |       |
| Mason Stain 6391 ..... | 1.6 % |

Yields a smooth transparent glaze that is great over carved or incised decoration on porcelain.

From Tom & Elaine Coleman,  
*Ceramics Monthly*, January 2003.



# Glaze defects

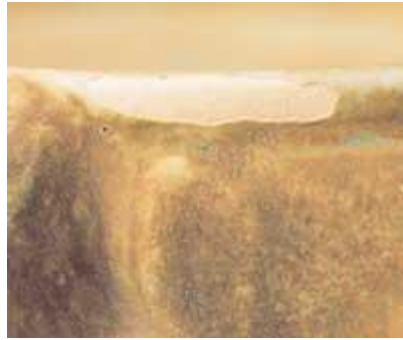


Crazing

Solving glaze defects: Crazing is the most common defect, and normally the easiest to correct. In both crazing and shivering the eradication of problems relies on matching the thermal expansion characteristics of both body and glaze. In practice, the most effective ways to correct crazing are:

1. increase the silica in glaze
2. decrease the feldspar in body
3. decrease any other material containing sodium or potassium in glaze
4. increase the alumina, i.e. the clay content, in glaze

## Shivering



Shivering is the most problematic of glaze defects, particularly in any functional ware, since shivers of glaze from the edges can drop into food or drink. These shivers are usually small, sharp, jagged-edged pieces that detach themselves from the body. It is the most dangerous of glaze faults. Shivering is the reverse of crazing, therefore the remedies are the opposite as well:

1. decrease the silica in body
2. increase the feldspar, especially sodium feldspar or nepheline syenite, or other alkaline bearing materials, in body

## Crawling



Crawling is caused by a high index of surface tension in the melting glaze. It is triggered by adhesion problems, often caused by bad application. It occurs where a glaze is excessively powdery and does not fully adhere to the surface of the clay. This can be alleviated by the addition of a small amount of gum to the glaze batch. Crawling is more common in matt glazes than in fluid ones; sometimes the problems of crawling can be reduced by the addition of a small amount of extra flux. Crawling can also occur when one glaze is applied over another, particularly if the first is allowed to dry out completely before the second application. Some fluxes, particularly zinc and magnesium, are likely to cause crawling when used in excess. Calcining all or part of the zinc can help this problem

## Pitting and Pinholing



These are the most annoying and difficult glaze flaws to cure. They can be caused by a badly controlled firing cycle, the glaze composition, or can originate with the body, particularly highly grogged clay bodies. The following remedies should be tried to cure pinholing or pitting:

1. lengthen the firing cycle
2. apply the glaze less thickly
3. add more flux to the glaze to make it more fluid
4. decrease the content of zinc or rutile in the glaze
5. where zinc is used, try calcining half or all of the zinc content
6. increase the maturing temperature of the glaze
7. hold the kiln at the glaze maturing temperature for a soaking period of up to two hours
8. cool the kiln slowly.

### 3. วัสดุเซรามิกขั้นสูง (Advanced ceramic materials)

# Alumina as structural ceramic

Properties of reactive grade alumina:

impurities  $\text{Na}_2\text{O}$  0.08wt%

surface area  $6.8\text{m}^2\text{g}^{-1}$

sintered density 3.92 (2h 1650°C)

bend strength 500 - 600 MPa

melting temperature

2050°C

sintering temperature

1550 - 1600°C

fracture toughness

4 - 4.5 MPa $\text{m}^{1/2}$

Applications: hip prostheses, cutting tools (zirconia-toughened)



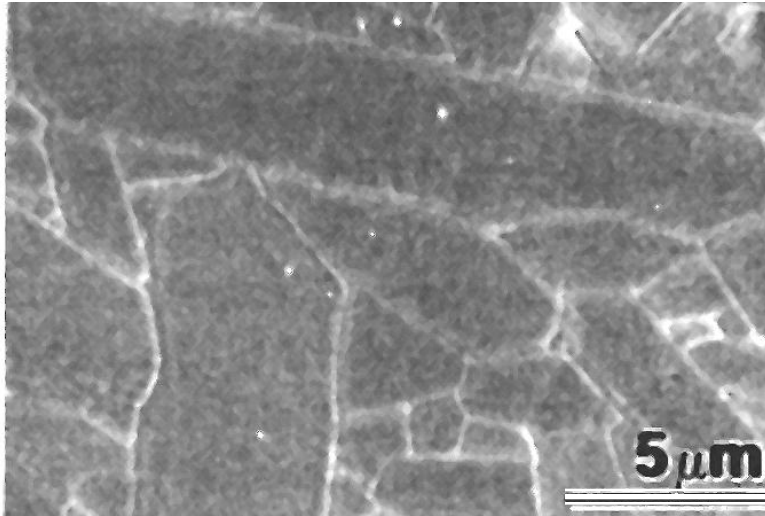
Triangular alumina-based cutting element used to machine metallic parts



Cutting elements made of alumina

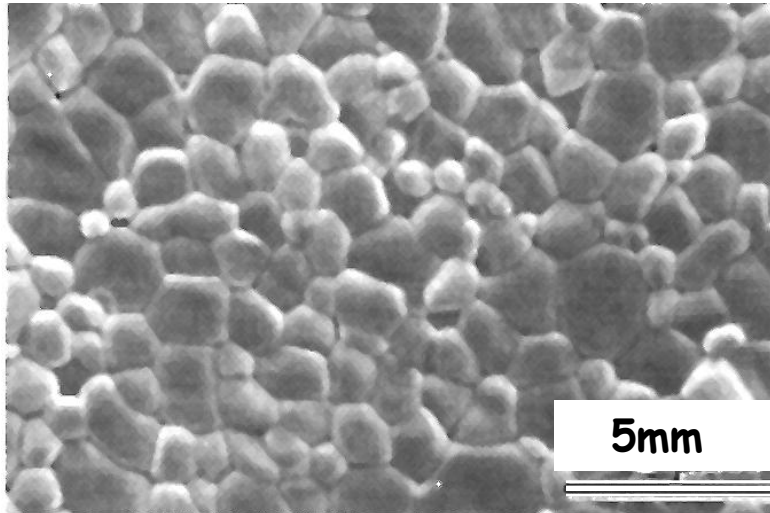
# Alumina: microstructure and strength I

Controlling the microstructure of alumina ceramics to enhance mechanical properties



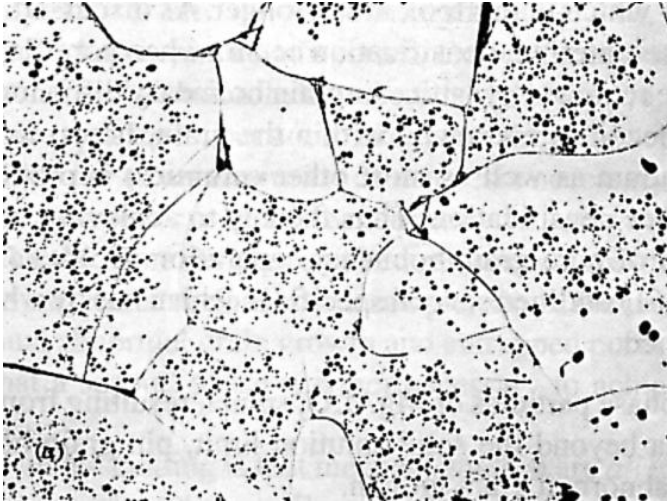
Dense hot pressed alumina without (top) and with addition of MgO (bottom) Grain growth is detrimental to the fracture strength of ceramics:

$$\sigma_f \propto d^{-1/2} \quad d: \text{grain diam.}$$



Doping alumina with MgO leads to the formation of precipitates of spinell along the grain boundaries, which lowers the grain boundary mobility. (Bennison et al., 1983)

## Alumina: microstructure and strength II

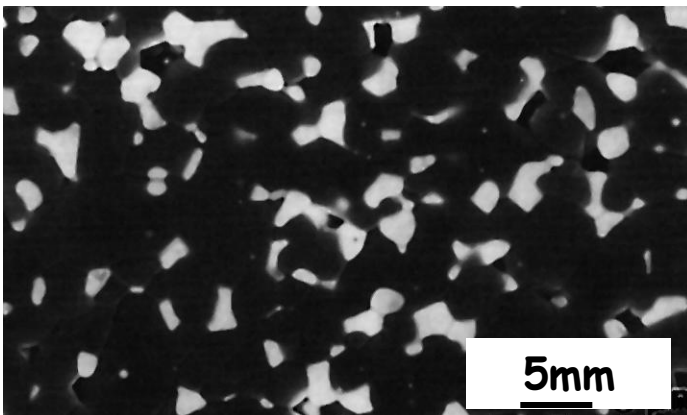


Porosity is detrimental to the mechanical strength:

$$\sigma_f = \sigma_0 \exp^{(-bp)}$$

$\sigma_0$ : strength at zero porosity  
 $b$ : constant.

Doping alumina with periclase reduces also the internal residual porosity. The picture (Geskovich et al. 500x) shows an alumina body sintered without dopant. There is a large number of entrapped pores. When sintered with a dopant, the reduced grain boundary mobility allows the filling of the pores when they are at the grain boundaries, whereas fast grain growth encloses the pores quickly into the interior of the grain, where it is difficult to eliminate them.



Pure alumina has a low fracture toughness. Mixing ca. 10% of zirconia (BSE image, zirconia: white) into the alumina doubles the fracture toughness.

# Pepper / Salt Grinder

## Processing

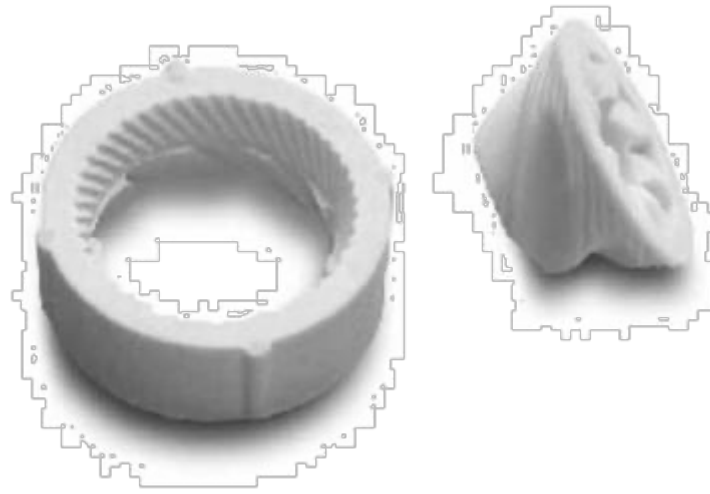
- Net-shape  
Injection Molding

## Properties

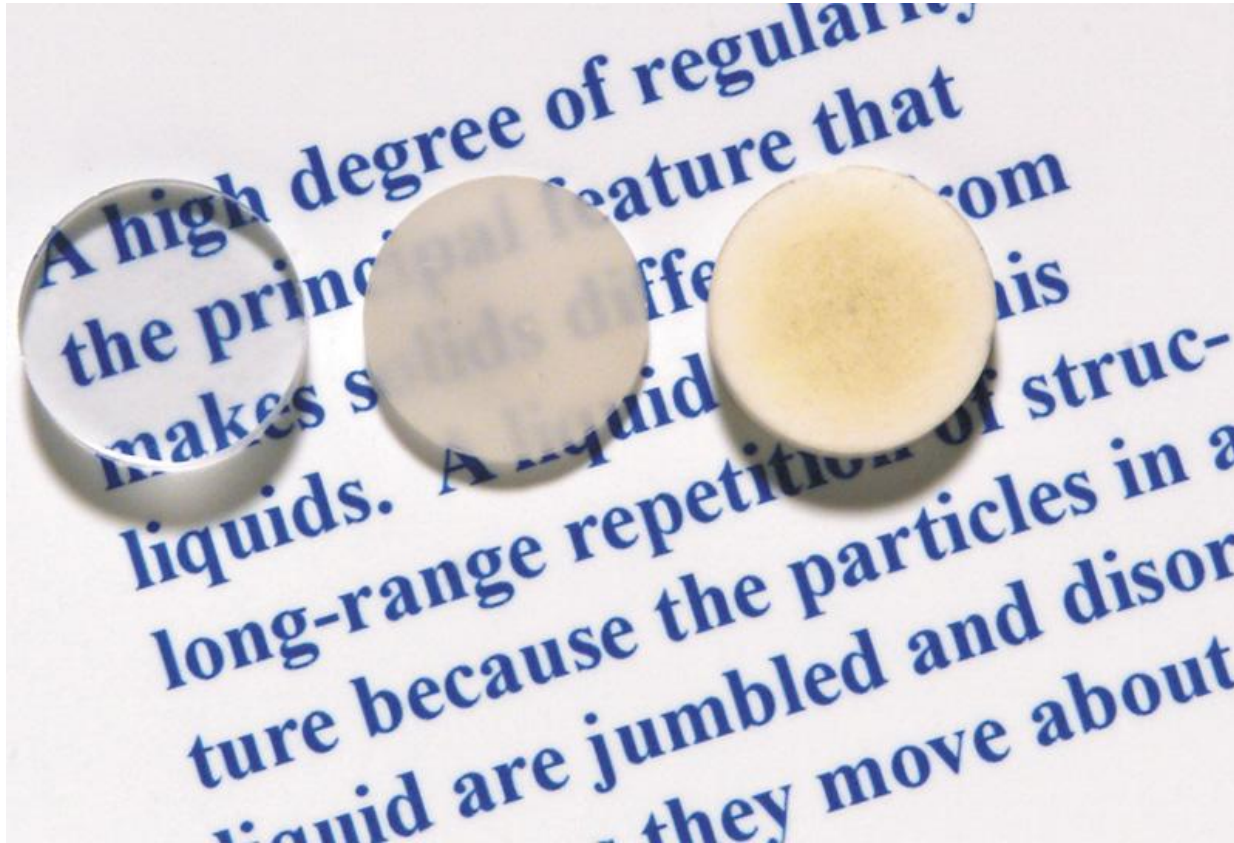
- High Hardness
- Resistance against NaCl

## Advantages

- No Corrosion
- Long lifetime
- Cheaper



# OPACITY AND TRANSLUCENCY IN ALUMINA



the light transmittance of three aluminum oxide specimens. From left to right: single-crystal material (sapphire), which is transparent; a polycrystalline and fully dense (nonporous) material, which is translucent; and a polycrystalline material that contains approximately 5% porosity, which is opaque.

# Zirconia as structural ceramic

Properties of partially stabilized zirconia:

dopant  $\text{Y}_2\text{O}_3$ ,  $\text{CaO}$  3 - 10wt%

sintered density  $6.05 \text{ gcm}^{-3}$

Young's modulus 170 - 210 GPa

Bend strength 400 - 700 MPa

melting temperature

2500°C

sintering temperature

1800°C

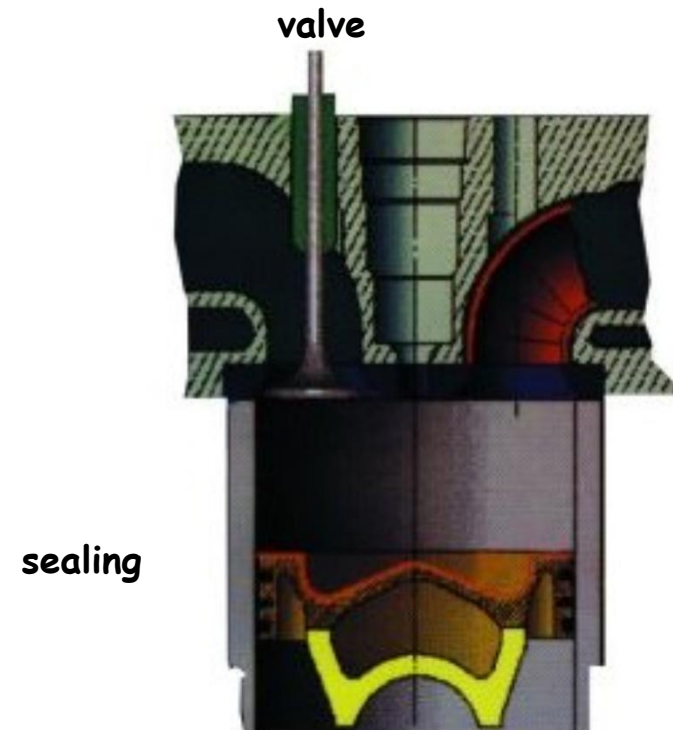
fracture toughness

6 - 20  $\text{MPam}^{1/2}$

Applications: die material in the metall industry, thermal barrier coatings, piston caps, cutting tools



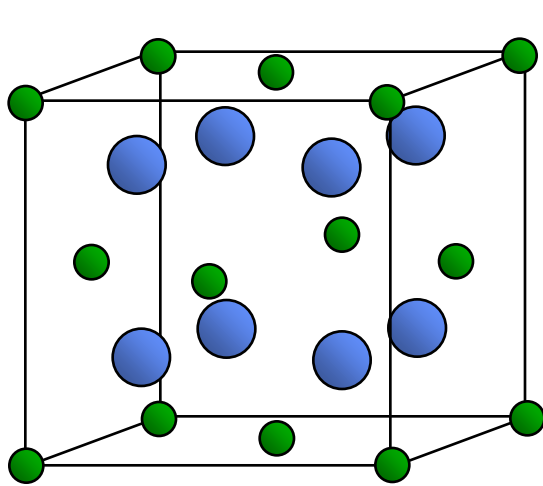
Piston parts (valves, sealings etc. made of stabilized zirconia.



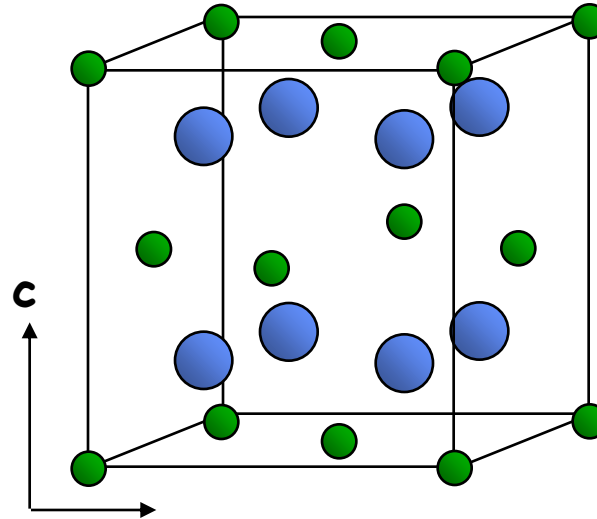
Schematic drawing of a piston.

# Polymorphs of $\text{ZrO}_2$

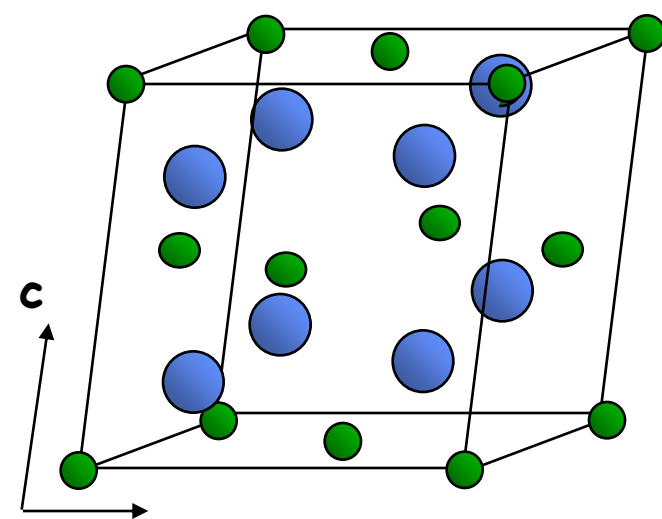
Schematic structures of the three zirconia polymorphs



cubic c-phase  
2370°C - 2680°C



tetragonal t-phase  
 $c/a = 1.02!$   
1240°C - 2370°C



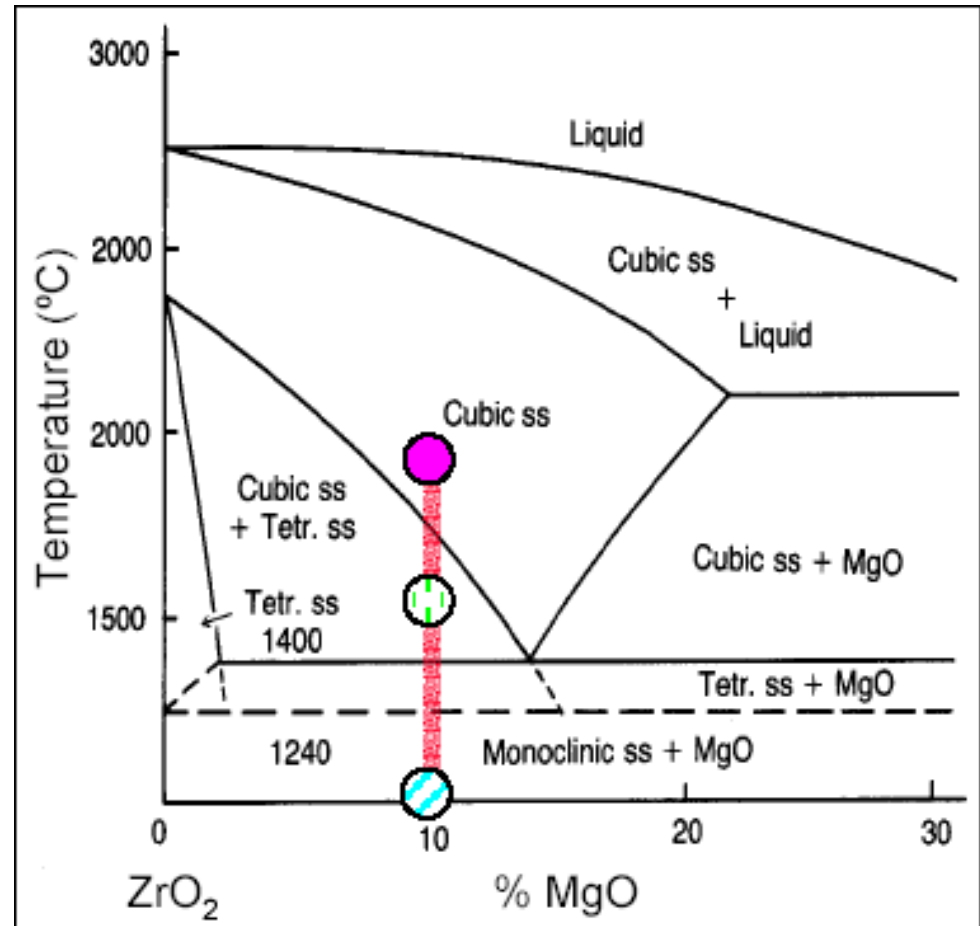
monoclinic m-phase  
< 1240°C

- The cubic phase can be *stabilized* by *doping* with  $\text{MgO}$ ,  $\text{CaO}$  or  $\text{Y}_2\text{O}_3$
- The tetragonal - monoclinic phase transformation involves a 4.7% volume increase.
- This volume increase is the basis for transformation toughening.

## Partially stabilized zirconia (PSZ)

### ~~Manufacturing of partially stabilized zirconia~~

- ~~• Add about 10% MgO~~
- ~~• Sinter in the cubic phase~~
- ~~• Lower temperature and heat treat (age) to nucleate small precipitates of t-phase~~
- ~~• These are growing below the critical size for t-m transformation~~
- ~~• Cool to room temperature~~
- ~~• Remaining c-phase has no time to transform~~



ZrO<sub>2</sub>-MgO phase diagram

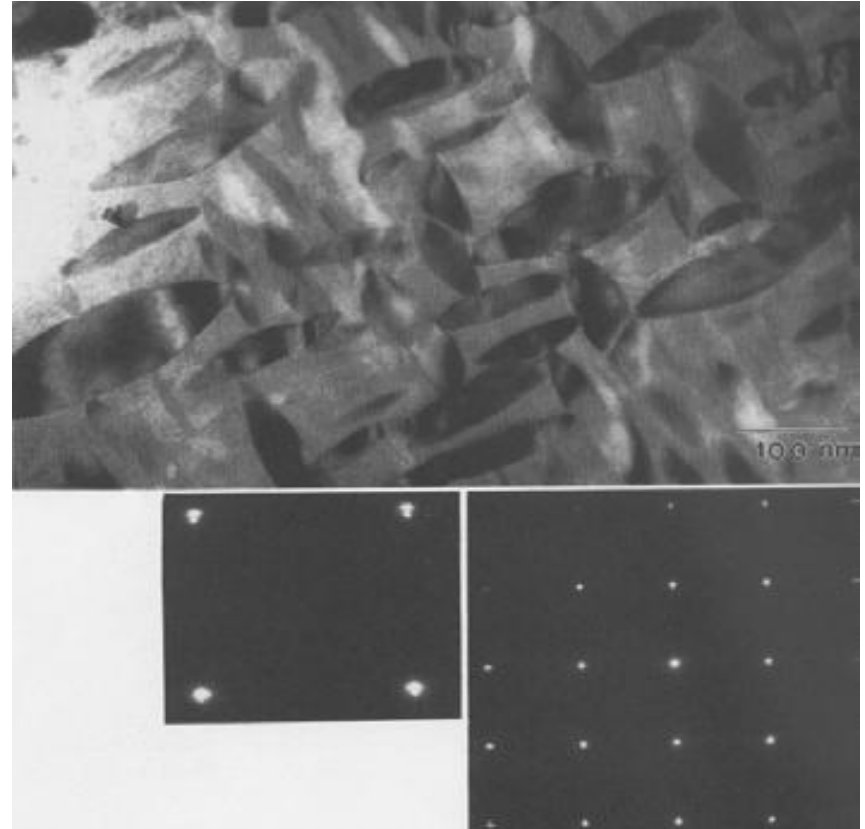
# Mg-PSZ Microstructures

After sintering at 1800°C an annealing stage at 1400°C is introduced:

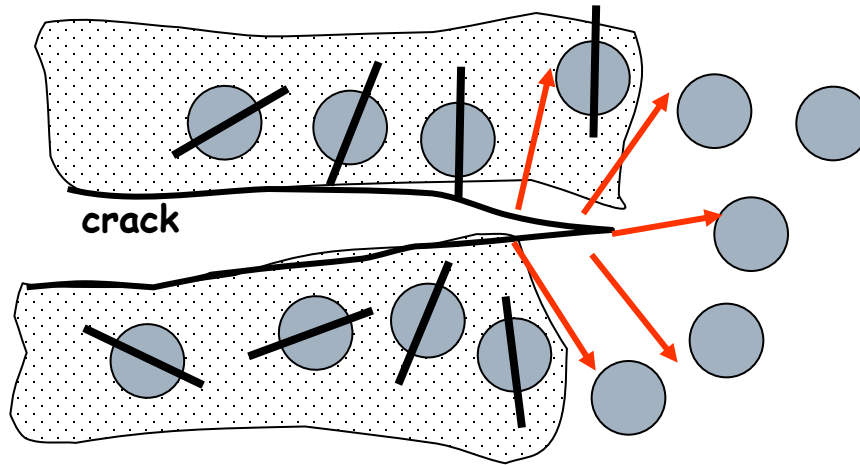
- After 4-5 hours tetragonal precipitates, grow by conventional diffusion processes as coherent spheroids along {001} cube planes

- Below a well defined critical size of about 200 nm the t-particles remain tetragonal down to room temperature

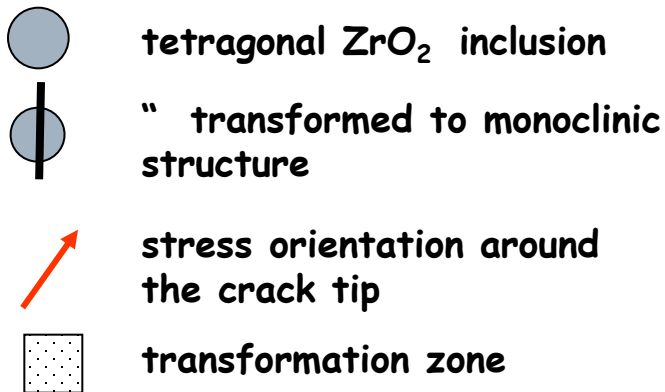
- Optimum microstructures contains about 25% - 30% by volume of tetragonal phase



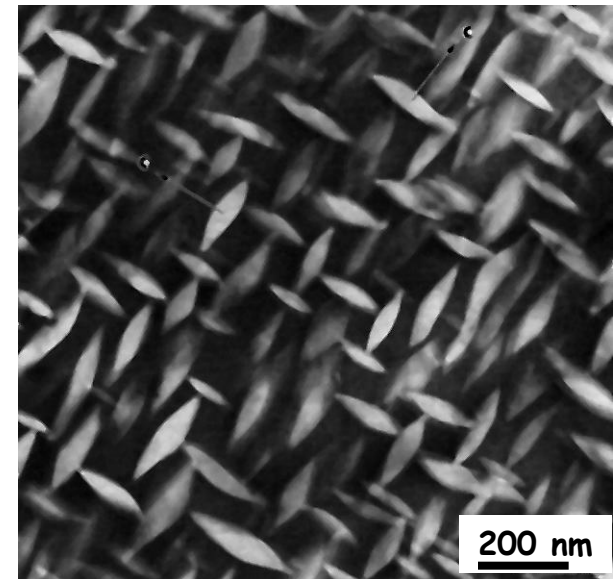
# Transformation toughening I



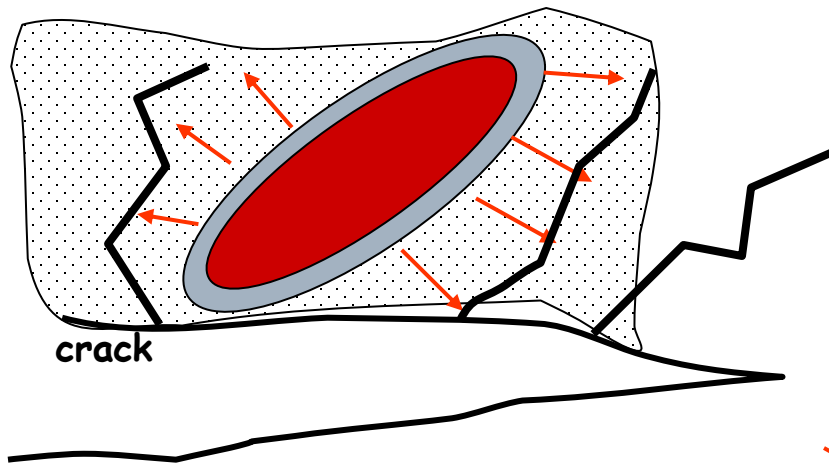
1. The stresses concentrated at the crack tip transform the surrounding tetragonal  $\text{ZrO}_2$  inclusions to the monoclinic polymorph. The transformation absorbs fracture energy and slows down crack propagation.



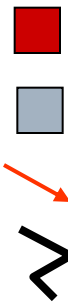
Lense-shaped tetragonal inclusions in a matrix (black) of cubic zirconia (A. Heuer).



# Transformation toughening II



2. Microcracking around the transformed inclusions: The volume stresses resulting from the tetragonal-monoclinic transformation delocalize also the stresses from the crack tip

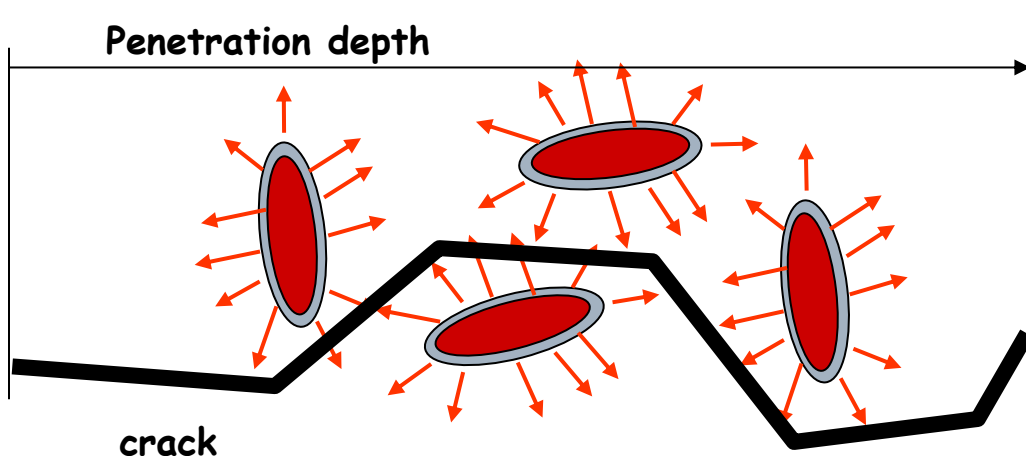


volume of the tetragonal zirconia inclusion

volume after transformation to monoclinic

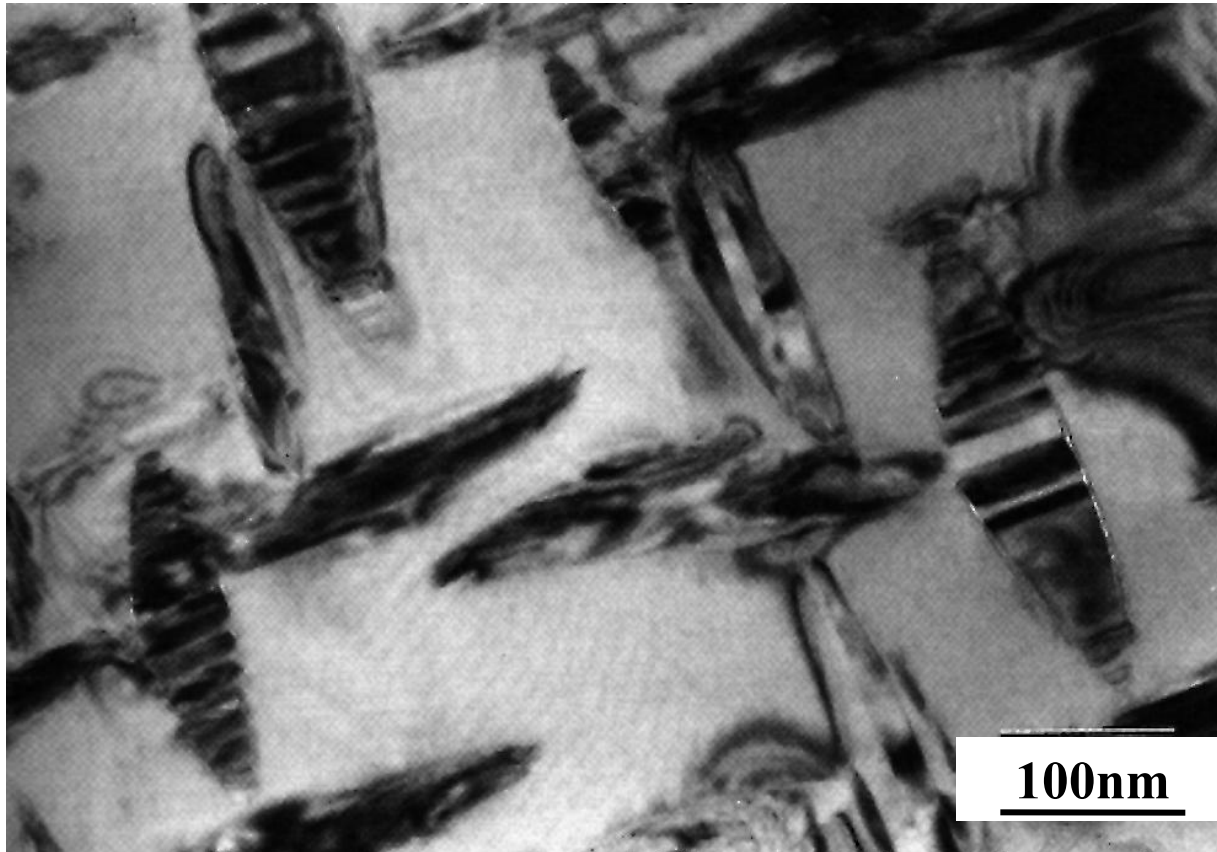
stresses due to the volume increase

microfracture due to the volume stresses



3. Crack deflection due to volume stresses: The deflection of cracks increases the crack surface. The stress release per unit penetration is, therefore, larger than for an inclusion free zirconia.

## Transformation toughening III



Initially tetragonal zirconia inclusion in a cubic zirconia matrix, which are completely transformed to the monoclinic structure. The bands within the inclusions are twin lamellae.